

3 2

2

4

5

6

3	13	15	2	7	8	10
0	1	2	3	4	5	6

2	3	13	15	7	8	10
0	1	2	3	4	5	6

key < arr[j]
 $13 < 15$
 $j--$
 $-1 \times$
 $j+1 = \text{Key}$
 $-1+1 = \text{Key}$
 $0 = \text{Key}$

Trace insertion sort for the following :-

34	5	67	8	9	10	15	18
0	1	2	3	4	5	6	7

$n = 8$

$i = 1, 2$

$i < n$

$1 < 8 \checkmark$

$2 < 8 \checkmark$

Key = arr[i]

$6+2$

$5 = 1$

$j = i-1 = 1-1$

$j = 1-1$

$j = 1-1$

> 0

Key < arr[j]

$67 < 34 \times$

$5 < 34 \checkmark$

$j--$

$j--$

$j--$

$j--$

$j--$

$j--$

$j--$

$j--$

$j--$

$j--$

$j--$

$j--$

$j--$

$j--$

$j--$

$j--$

i j Key

1 0-1 5

5	34	67	8	9	10	15	18
0	1	2	3	4	5	6	7

i j Key

2 1 0-1 67

5	34	67	8	9	10	15	18
0	1	2	3	4	5	6	7

i j Key
3 21 8

5	8	34	67	9	10	15	18
0	1	2	3	4	5	6	7

① ② Key i j

5	8	9	34	67	10	15	18
0	1	2	3	4	5	6	7

i j Key
5 432 10

5	8	9	10	34	67	15	18
0	1	2	3	4	5	6	7

i j Key
6 843 15

5	8	9	10	15	34	64	18
0	1	2	3	4	5	6	7

i j Key
7 654 18

5	8	9	10	15	18	34	64
0	1	2	3	4	5	6	7

i = 3
j < 0
3 < 8 ✓

Key = arr[i]
8 = 3

i = i - 1
j = 3 - 1
j = 2

Key < arr[j]
8 < arr[2]

8 < 67 ✓

8 < 34 ✓